Trainings

General Information

All maintenance procedures listed for previous intervals **must** also be performed.

Listed below are the section numbers that contain specific instructions for performing the maintenance.

Daily - Maintenance Procedures (Section 3)

- Air Cleaner Element - Check
- Air Cleaner Precleaner - Check
- Air Cleaner Restriction - Check
- Air Intake Hoses, Pipes, and Clamps - Check
- Air Tanks and Reservoirs - Drain
- Coolant Level - Check
- Drive Belts - Check
- Fuel-Water Separator - Drain
- Lubricating Oil Level - Check

Every 250 Hours or 6 Months - Maintenance Procedures (Section 4)

- Belt Tension - Check
- Belt Tensioner - Check
- Fan Hub, Belt-Driven - Check
- Coolant Filter - Change
- Air Cleaner Element - Change
- Cooling Fan - Check
- Crankcase Breather Tube - Check
- Engine Wiring - Check
- Fuel Filters - Change[1]
- Lubricating Oil and Oil Filter - Change[2]
- Coolant Filter - Change[3]
- Supplemental Coolant Additive - Check
- Air Compressor Air Cleaner Element- Check[4]
- Water Pump Weep Hole - Check

Every 1500 Hours or 1 Year - Maintenance Procedures (Section 5)

- Batteries - Check
- Cold Start Aids (Seasonal) - Check
- Crankcase Breather (External) - Clean
- Engine Mounting Bolts - Check
- Hoses, Engine - Replace

- Steam Clean Engine - Clean

Every 6000 Hours or 2 Years - Maintenance Procedures (Section 6)

- Cooling System - Clean
- Fan Hub, Belt-Driven - Check
- Fan Idler Pulley Assembly - Check
- Vibration Damper - Check
- Water Pump - Check
- Air Compressor Discharge Lines - Check
- Overhead Set - Check
- Air Intake Manifold Heater - Check

Every 10,000 Hours - Maintenance Procedures (Section 7)

- Fuel Injectors (All applications with Mechanically Actuated Fuel Injectors) - Replace
- Turbocharger - Check[5]

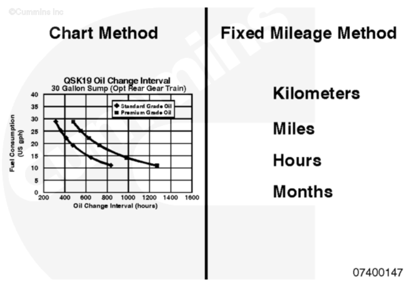
Notes:

1. The replacement of the fuel filters can be extended to 500 hours, as long as the fuel inlet restriction does **not** exceed the maximum specified in Section V.
2. The lubricating oil and filter change intervals can be extended, based on the engine's horsepower rating, oil pan capacity, and fuel consumption rate. Reference the Oil Drain Intervals information in this section for specific details.
3. The lubricating oil and filter change intervals can be extended, based on the engine's horsepower rating, oil pan capacity, and fuel consumption rate. Reference the Oil Drain Intervals information in this section for specific details.
4. Cummins Inc. recommends the use of dry-type air compressor air cleaners.
5. For engines operating with a high altitude calibration, the turbocharger **must** be replaced at 10,000 hours or engine half life, whichever comes first.

Oil Drain Intervals

⚠ CAUTION ⚠

The use of synthetic base oil does not justify extended oil change intervals. Extended oil change intervals can decrease engine life due to factors such as corrosion, deposits, and wear.



⚠ CAUTION ⚠

Do not extend oil change intervals beyond the maximum point shown when fuel consumption rates are less than the minimum shown. Doing so can result in shortened engine life

due to oil degradation processes associated with lightly loaded engines.

Do **not** extend oil and filter change intervals beyond those specified configurations in the maintenance schedule, unless the chart method is used. See the oil interval charts in this section. Reference the following service bulletins:

- Cummins® Engine Oil and Oil Analysis Recommendations, Bulletin 3810340 (/qs3/pubsys2/xml/en/bulletin/3810340.html).
- Oil Analysis Techniques for High Horsepower Diesel Engines, Bulletin 4022060 (/qs3/pubsys2/xml/en/bulletin/4022060.html).
- Statistical Analysis of Oil Sample Lead Readings on High Horsepower Engines, Bulletin 2883452 (/qs3/pubsys2/xml/en/bulletin/2883452.html).

There are two recommended methods to determine the proper oil and filter change interval:

- Fixed Hour Method (based on fixed hours or months; which ever occurs first)
- Chart Method (based on known fuel consumption rates, coupled with oil analysis program).

The chart method is recommended to provide the lowest total cost of operation, while still protecting the engine.

Fixed Mileage Method				
	56 liters [14.8 gal] pan		**135 liters [35.6 gal] pan**	
QSK23 Engine Rating	CF-4, CES (Cummins Engineering Standard) 20075 Grade Oil	CH-4, CI-4, CJ-4, CES 20071, CES 20076 Grade Oil	CF-4, CES 20075 Grade Oil	CH-4, CI-4, CJ-4, CES 20071, CES 20076 Grade Oil
Up to 800 hp	250 hours	250 hours	250 hours	500 hours
Above 800 hp	250 hours	250 hours	250 hours	500 hours

Chart Method:

The information listed below is required when using the chart method to determine the correct oil and filter change interval for the engine.

- Oil system capacity (sump plus any remote tank volume)
- Average fuel consumption rate

- Use engine oil that complied to the standards listed in Service Bulletin 3810340 (/qs3/pubsys2/xml/en/bulletin/3810340.html).
- Oil analysis to demonstrate the chart method interval is acceptable. Reference the following service bulletins:
    - Cummins® Engine Oil and Oil Analysis Recommendations, Bulletin 3810340. (/qs3/pubsys2/xml/en/bulletin/3810340.html)
    - Oil Analysis Techniques for High Horsepower Diesel Engines, Bulletin 4022060. (/qs3/pubsys2/xml/en/bulletin/4022060.html)
    - Statistical Analysis of Oil Sample Lead Readings on High Horsepower Engines, Bulletin 2883452. (/qs3/pubsys2/xml/en/bulletin/2883452.html)

System capacity can be determined by knowing the volume of the oil required to touch the high-level mark on the dipstick and volume of any remote oil tanks on the machine in which oil is continuously circulated. Oil sump capacities are listed in the operation and maintenance manuals for all Cummins® engines. It is imperative that the total system volume be known when using the chart method. Use the following procedure for information on the oil sump capacity. Refer to Procedure 018-017 in Section V (/qs3/pubsys2/xml/en/procedures/89/89-018-017.html) for system capacity. If the machine is equipped with an oil reserve system with a reservoir remote from the engine oil sump, the reservoir volume **must** be added to the engine sump volume to determine the total system capacity. This is **only** true for remote tanks in which the oil is continuously circulated.

When oil is changed on an engine with a remote oil tank in which oil is continuously circulated, the oil in the tank **must** be changed in addition to the oil in the engine sump.

To use the chart method effectively, accurate fuel consumption records **must** be kept and maintained. Fuel consumption information **must** be in liters per hour or gallons per hour units to use the charts. Fuel consumption rates can change over time because of an increasing or decreasing engine duty cycle. Accurate records are essential in determining average fuel consumption during a given oil change interval.

The charts are categorized and labeled by the system capacity for a specific engine family. Select the chart representing the correct oil system capacity for engine model in question. The left vertical axis of the chart represents fuel consumption in U.S. gallons per hour. The chart method shows both gallons per hour and liters per hour. Determine the intersection point of the fuel consumption rate on the applicable interval curve by drawing a horizontal line. From this point, draw a vertical line down until it intersects with the horizontal axis representing hours. This point represents the acceptable oil change interval.

Note : Before using the charts below it is important to note the grade of oil being used. Reference Table 1. This can affect which chart can be used.

Table 1: Engine Oil Grades			
North America Classification	**International Classification**	**CES**	**Description**
API[1] CD API CE	ACEA[2] E1	Standard obsolete. These oils are no longer recommend for use in Cummins® engines.	Do **not** use the Chart Method if using these oils.
API[1] CG-4/SH			
API CF-4/SG	ACEA[2] E2	CES 20075[7]	
	ACEA2 E3		
API[1] CH-4[4] /SJ	JAMA[3] DH-1 ACEA[2] E5[5]	CES 20071	Engines using these oil grades can use the Chart Method
		CES 20076 CES 20077	
API[1] CI-4	ACEA2 E7	CES 20078	
API[1] CJ-4[8]	ACEA[2] E9 JAMA3 DH-2	CES 20081	

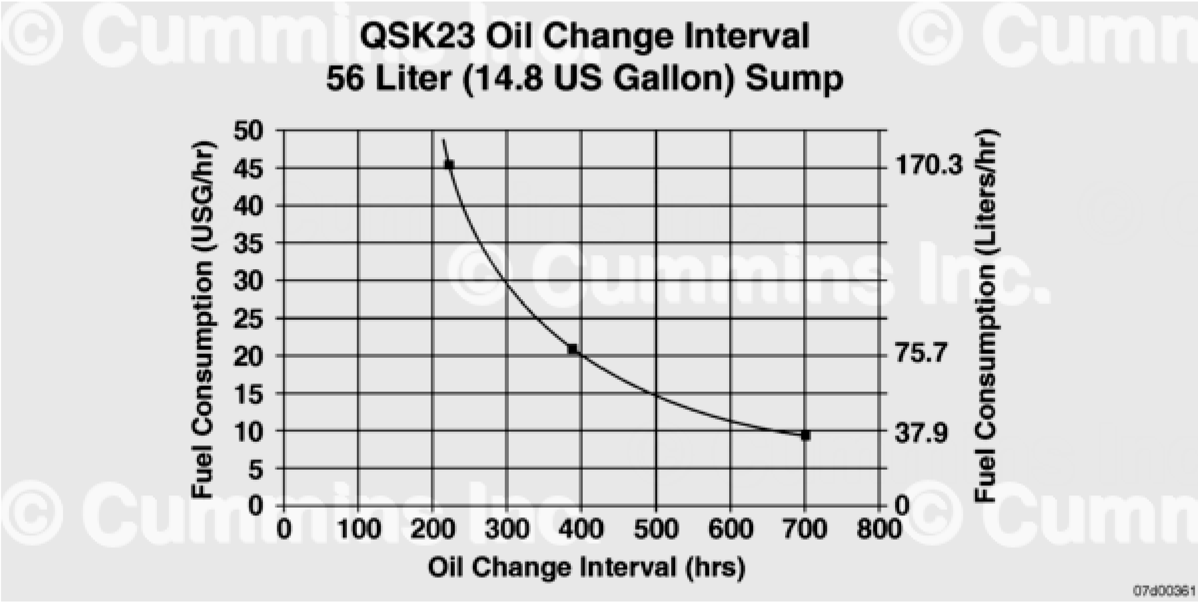
1. API - American Petroleum Institute.
2. ACEA - Association des Constructeurs European d' Association. ACEA - Association des Constructeurs European d' Association.
3. JAMA - Japanese Automobile Manufacturers Association. JAMA - Japanese Automobile Manufacturers Association.
4. CES 20076 adds the requirement of a 300 hour Cummins® M11 test to API CH-4. CES 20076 adds the requirement of a 300 hour Cummins® M11 test to API CH-4.
5. CES 20077 adds the requirement of a 300 hour test to ACEA E5. CES 20077 adds the requirement of a 300 hour test to ACEA E5.
6. Use of oils with **only** these designations poses an undue risk of engine damage for engines designed to use more advanced oils, even when drastically shortened oil change intervals are followed.
7. CES 20075 CF-4/SH and E-3 oils can be used in areas where none of the recommended oils are available, but the oil drain interval **must** be reduced. Reference the appropriate maintenance schedule. Use of oils with **only** these designations poses an undue risk of engine damage for engines designed to use more advanced oils, even when drastically shortened oil change intervals are followed.
8. API CJ-4 oil should **only** be used on engines running with Ultra Low Sulphur Diesel fuel.

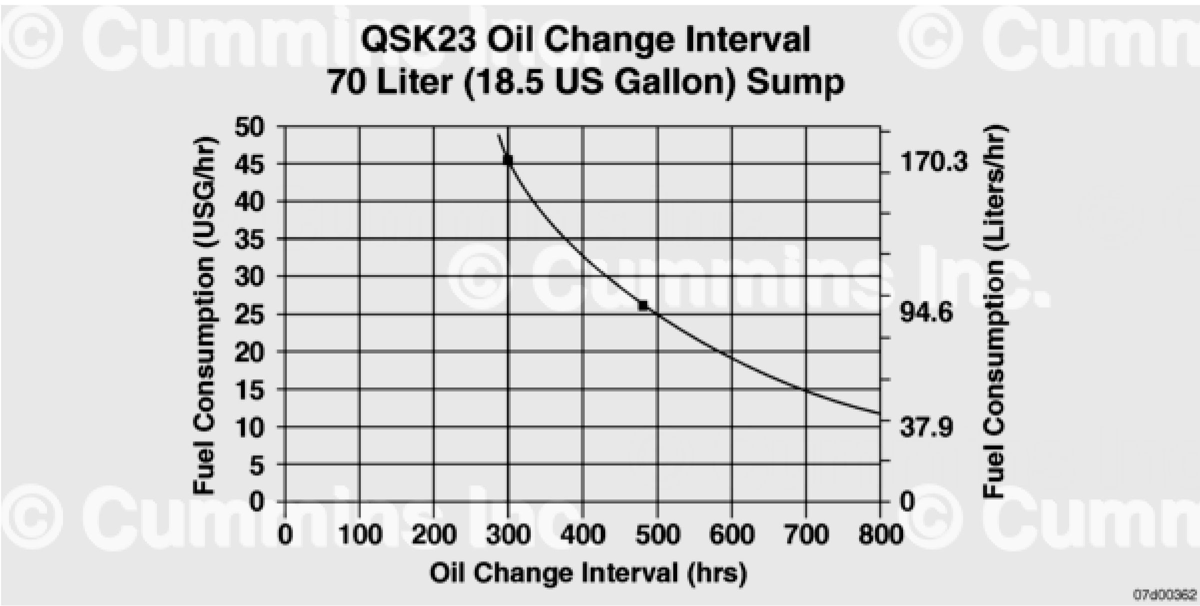
⚠ CAUTION ⚠

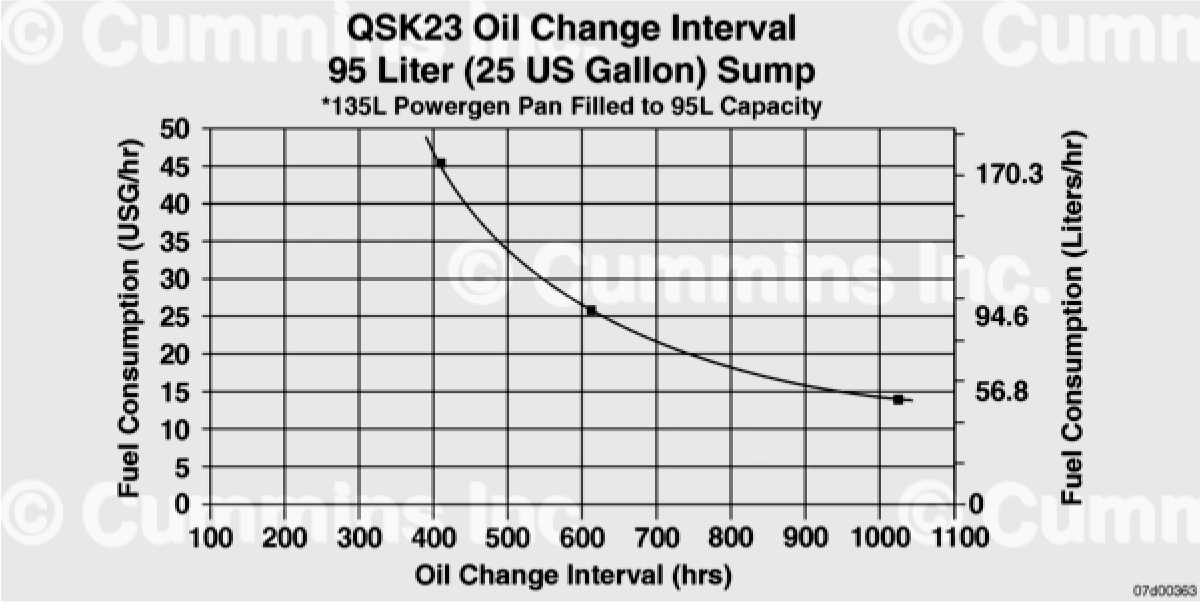
Inaccurate fuel consumption records can result in incorrect oil change intervals that can result in premature engine wear and/or deposits.

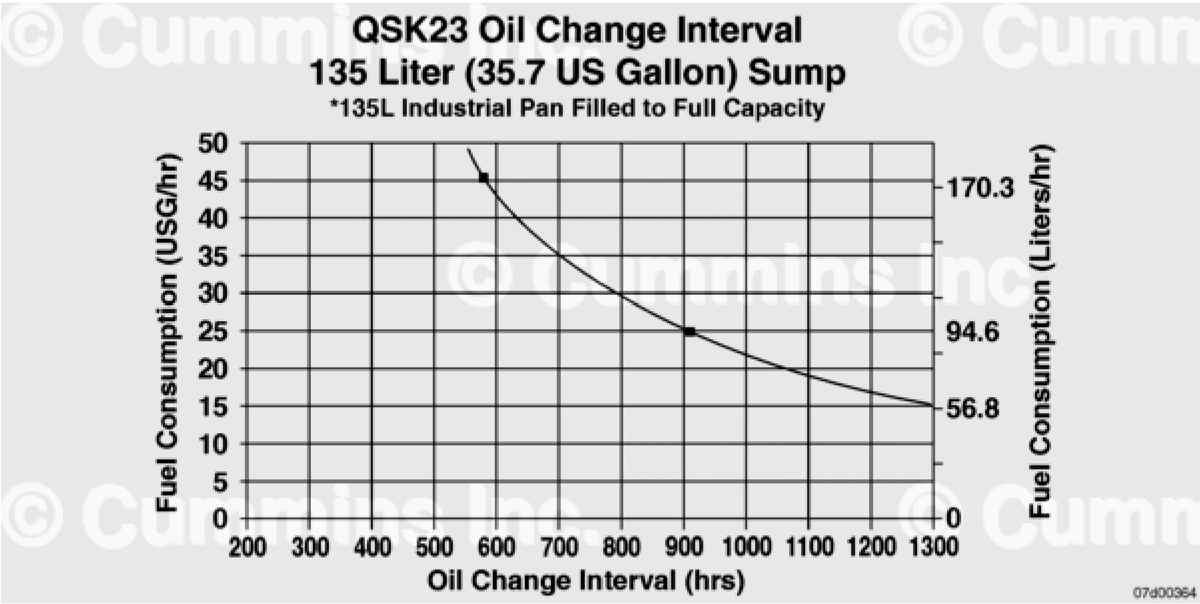
⚠ CAUTION ⚠

Do not extend oil change intervals beyond the maximum point shown when fuel consumption rates are less than the minimum shown. Doing so can result in shortened engine life due to oil degradation processes associated with lightly loaded engines.







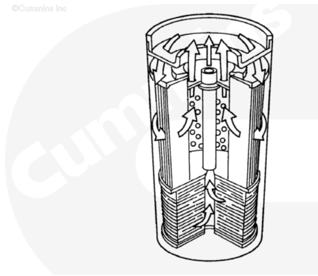


To read the chart:

Assume the engine has a 14.8-gallon sump capacity and the engine consumes 15 gallons of fuel per hour.

- Select the chart titled 14.8 gallons.
- The left vertical axis of the chart represents fuel consumption in gallons per hour.
- Determine the location of 15 gallons on the left vertical axis and draw a horizontal line across the chart until it intersects the curve.
- At the point of intersection, draw a straight vertical line down to the bottom of the graph.
- The point at which the vertical line intersects the bottom horizontal axis of the graph indicates the oil change interval (approximately 500 hours in this example).

Lubricating Oil Filter



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Premium oil filters should be used in conjunction with premium-grade oils when using the premium-grade oil curves to determine oil change intervals. Refer to Section V for lubricating oil filter specifications.

Premium filters contain synthetic media materials to provide more efficient filtration for the entire service life and extend the media life over conventional cellulose media.

Premium combination filters are made with synthetic media and have StrataPore™ designation on the outside of the filter.

StrataPore™ filters have the efficiency, capacity, and strength needed for extended service intervals.

Fleetguard® DCA4 Service Filters and Liquid Precharge

Maintenance Intervals for Cooling Systems up to 1514 Liters [400 Gallons]										
Service Interval [Hours]	**System Size in Liters [Gallons]**									
	79 to 144 [21 to 30]	117 to 189 to [31 to 50]	193 to 284 [51 to 74]	288 to 378 [76 to 100]	382 to 568 [101 to 150]	572 to 757 [151 to 200]	761 to 946 [201 to 250]	950 to 1135 [251 to 300]	1139 to 1125 [301 to 350]	1329 to 1574 [351 to 400]
751 to 1000	25	50	80	100	250	200	250	300	350	400
501 to 750	20	35	60	75	220	150	190	225	260	300
251 to 500	15	25	40	50	75	100	125	150	175	200
0 to 250	10	16	20	25	40	50	65	75	90	100
Install service filter(s) and/or liquid containing number of supplemental coolant additive (SCA) units above.										

Last Modified: 30-Apr-2024